

Constructing the Source Code of the Universe: From the Tachyon Sea to a Theory of Everything (TOE)

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Abstract: *This paper presents a dynamic model of cosmic evolution that covers the phase before the Big Bang through a thought experiment — our universe arises from the local decoherence of a fully entangled tachyon sea. The model is logically self-consistent, elegant, and contains no singularity. The evolution after the formation of our universe is completely consistent with the cosmic inflation model. The entire universe exhibits a cyclic, renormalizing evolution process, naturally unifying relativity and quantum mechanics. In the evolution of tachyons within an isolated system, the tachyon hypothesis provides a concrete and unified physical picture for string theory and loop quantum gravity, reflecting a duality under the physical perspective: the internal view (string theory) is that of a tachyonic field with imaginary mass and high energy; the external view (loop quantum gravity) is a “geometric envelope” of high curvature, negative energy, and violent oscillation — namely, excited gravitons. Since the total energy $E_{\text{total}}=0$, gravitons and tachyons are tightly coupled inside space, while externally only changes in geometry and volume are manifested, with vibrational information encoded on the spatial surface. The model intuitively combines the physical pictures of string theory (high-dimensional vibrational modes) and loop quantum gravity (background-independent geometric quantization), pointing toward a more fundamental “quantum information - geometry” unified theory. It provides a self-consistent physical picture and mechanism for the following frontier problems: where spacetime comes from, the black hole information paradox, what gravity is, dark matter, vacuum energy, why the dark energy (cosmological constant) is small, ...*

1 An Interpretation of the Emergence of Space and Time

In recent years, the AdS/CFT correspondence and its core corollaries—such as the Ryu-Takayanagi formula and the entanglement wedge construction—have shown that quantum entanglement in the boundary field theory can fully encode the geometric information of the higher-dimensional gravitational spacetime^[1, 2]. The structural constraint that the entanglement wedge must always contain the causal wedge further suggests that entanglement is not merely a passive participant in the duality but is the fundamental source of spatial geometry^[3]. Holographic duality is mostly restricted to static or asymptotically AdS backgrounds. To understand our universe—an expanding spacetime with macroscopic twistability and microscopic discreteness—it is necessary to combine entanglement with decoherence processes. The following are the core inferences of spacetime emergence based on this theory.

1.1 Emergence of Space: From Entanglement Network to Classical Geometry

Space is not a pre-existing continuous manifold but an effective geometry emergent from a Planck-scale quantum entanglement network. Its core mechanisms are as follows:

Entanglement defines proximity: In holographic duality (AdS/CFT), the mutual information $I(A:B)=S_A+S_B-S_{AB}$ between two subregions A and B in the boundary quantum field theory is

proportional to the minimal surface area of the intersection of their corresponding entanglement wedges [1]. This implies that the spatial “distance” between two points is determined by the strength of quantum entanglement between them. For the global pure state $|\Psi\rangle_{\text{univ}}$, the entanglement strength $E_{ij}=S(\rho_i)+S(\rho_j)-S(\rho_{ij})$ between any two Planckian units i, j is monotonically anti-correlated with the effective geometric distance $d(i,j)\sim f(1/E_{ij})$. Strong entanglement corresponds to proximity; weak or zero entanglement corresponds to remoteness or even disconnected components.

Quantum network model of spacetime: At the Planck scale $l_P\sim 10^{-35}\text{m}$, spacetime is hypothesized to be a network composed of qubits (or more general quantum degrees of freedom). Each node represents a Planck volume unit, and edges between nodes represent entanglement correlations. The total entanglement structure of the entire network uniquely defines the intrinsic geometry of the universe. Continuity and smoothness of space are emergent results of the average over large-scale entanglement.

Unification of macroscopic twistability and microscopic discreteness: The twisting of spacetime described by the classical Einstein field equations (already confirmed by gravitational wave experiments), as well as the inevitable quantum nature at the Planck scale (e.g., the discrete area spectrum in loop quantum gravity) and “viscosity” (the pinning effect of energy-momentum changes on the spacetime structure), can all be regarded as macroscopic manifestations of the collective response of the underlying quantum entanglement network. The inhomogeneous distribution of the entanglement network directly gives rise to curvature; the response delay and dissipation of the network manifest as viscosity [5].

Macroscopic twistability and the nature of gravitational waves: The Einstein equations can be derived from the first-order variation of entanglement: for a holographic boundary subregion A , the variation of entanglement entropy δS_A with respect to shape deformation is related to the energy-momentum tensor via the Bianchi identity [5]. This means that macroscopic curvature corresponds to inhomogeneous distribution patterns in the entanglement network; gravitational waves are the propagation of collective excitation modes of the network, which, upon quantization, become gravitons.

Cosmic expansion = increase in the number of spatial quanta: In standard cosmology, expansion is described by the growth of the scale factor $a(t)$. However, in the discrete quantum network picture, expansion should be reinterpreted as an increase in the number of quanta: the total volume of the universe is $V(t)=N(t)\cdot V_0$, where V_0 is the size of a single Planck volume (the fundamental quantum) and remains constant; $N(t)$ grows with time. Expansion does not stretch the fundamental units but adds network nodes and entanglement edges. The total entanglement entropy $S_{\text{total}}(t)\propto N(t)$ increases with expansion and always satisfies the holographic entropy bound $S\leq A/(4l_P^2)$ [2].

According to the theory presented here, the stability of classical space originates from “entanglement crust” and continuous decoherence: the space produced by quantum entanglement is inherently unstable, but the universe, as an open system, continuously couples to its environment. Decoherence constantly refreshes the selection of geometric configurations, thereby locking in a definite macroscopic background. An isolated quantum network is in a superposition of multiple geometries; after coupling with the environment, decoherence suppresses the coherence between different geometries, causing collapse to a classical spacetime. A large number of microscopic entanglement modes then freeze into a stable classical metric, i.e., the “entanglement crust”. Holographic duality shows that introducing noise (simulating decoherence) into a boundary CFT

subregion reduces the recoverable region on the gravity side from the entanglement wedge to the causal wedge [4], indicating that decoherence indeed causes the loss of some geometric information, and what remains is the classically accessible spacetime.

1.2 Emergence of Time: Information Recording/Updating within a Reference Frame

Unitary evolution and microscopic reversibility: In an isolated quantum system, the dynamics of quantum entanglement are described by the unitary transformation $U(t)=e^{-iHt}$. This process preserves entropy and is in principle reversible. Therefore, time-reversal symmetry holds at the microscopic level, with no distinction between past and future. This is the starting point for the emergence of time — the arrow of time is not an inherent property of fundamental laws.

Decoherence generates event records: When the network couples to the environment (e.g., all high-energy degrees of freedom not explicitly monitored), environment-induced decoherence diagonalizes the system's density matrix and suppresses quantum interference between different geometric configurations. Each time the system interacts with the environment, a copy of the system's state is left in the environment, forming an indelible “event record.” The accumulation order of these records defines the direction of time.

Non-unitary mapping and the arrow of time: Classical time (the irreversible, directed arrow of time) does not come from microscopic unitary evolution; however, unitary evolution provides the basis for microscopic reversibility, while the irreversibility of time originates from information recording during decoherence. Formally, the decoherence process of a subsystem is described by a completely positive and trace-preserving map $\varepsilon(\rho_s) = \sum_k E_k \rho_s E_k^\dagger$, where the Kraus operators E_k are non-unitary. The discrete time step is written as $\rho_s^{t+1} = \varepsilon_t[\rho_s^t]$. Since tracing out the environment is irreversible, the composition order of these maps is non-commutative, thereby giving rise to the thermodynamic arrow of time.

Time as the order of information records: Therefore, the essence of time is the unidirectional accumulation of environmental memory. Without decoherence, there is no classical time; without an environment, there is no definite past or future. The irreversibility of **classical time = accumulation/updating of information records**: In the classical universe, every macroscopic event (e.g., stellar collisions, particle detections) leaves behind a large number of environmental memories (photons, thermal radiation, etc.). The unidirectional increase of these memories forms an irreversible causal order, which is the arrow of time we experience. Time is not a fundamental background stage, but rather a “running log” generated by the continuous “updating” of quantum information within the cosmic system, readable by internal observers. The universe is a dynamic quantum information network. The “present” is the consensus, unalterable log record within the network; “time flow” is the continuous updating of the system's state due to irreversible interactions (decoherence) between quantum information and the environment.

1.3 Phenomenological Interpretations

In this framework, the universe is viewed as an open quantum system, its “global log” consisting of a sequence of irreversible update events; the subjective time of each observer corresponds to the frequency and sequence with which they synchronize with the global log. The following reinterprets five key phenomena using this framework.

Twin Paradox: The “Relative Log” of Macroscopic Time and Its Resynchronization

The younger brother remains on Earth — Earth continuously interacts with the cosmic background radiation, forming a stable global information flow. Every physiological change and atomic clock oscillation of the brother is aligned in real time with this “cosmic master log”, so his time record is continuous and dense. The elder brother travels away at high speed; the spaceship becomes a quasi-closed “local information bubble”. Due to the Doppler shift and relativistic spacetime structure, the rate of information exchange between the spaceship and the Earth environment is greatly reduced (i.e., time dilation). During the cruising phase, the elder brother’s personal log evolves independently, updates slowly, and becomes desynchronized from the master log. The key is the turnaround and return: the spaceship must forcibly break its original inertial connection and undergo a violent information resynchronization with the Earth environment. This instantaneous “log merger” allows the elder brother to receive, in an extremely short time (subjectively very brief), the long history of updates accumulated by Earth. Consequently, the elder brother’s total log entries are far fewer than the younger brother’s, and upon reunion, the younger brother is older. Conclusion: the paradox arises from differences in the pace of information updates between inertial frames. The irreversible synchronization during the turnaround breaks the symmetry, providing an intuitive explanation without invoking acceleration details.

The “Timelessness” of Photons and Light-Speed Travelers: Zero Update Means Zero Proper Time

Special relativity predicts that a particle moving at the speed of light (e.g., a photon) has zero proper time. This means that from the photon’s perspective, the creation and annihilation of the universe occur simultaneously. In the information-update interpretation, this counterintuitive conclusion receives a direct and natural explanation. The source of updates is interaction: the passage of time corresponds to continuous information exchange between the system and its environment. For a freely propagating photon in vacuum, its evolution is described by the unitary operator $U(t)=e^{-iHt}$, which exchanges no information with the environment; the photon’s polarization, momentum, and other degrees of freedom remain unchanged, and no event occurs that can be externally recorded. Since no log entry is written, from the photon’s “subjective perspective” the number of time steps it experiences is zero. By extension, any entity moving at the speed of light — whether an elementary particle or a hypothetical “light-speed spaceship” — must satisfy the condition of zero information exchange with the environment. The moment one attempts to exchange information with it (e.g., send a signal), the entity no longer remains at light speed. Hence, “timelessness” is a universal feature of all light-speed movers, rooted physically in the absence of information updates rather than merely kinematic relations.

Wheeler’s Delayed-Choice Experiment: Postponed Log Submission and Post-Measurement Causal Fixation

Wheeler’s delayed-choice experiment reveals a puzzling feature of quantum mechanics: an observer can decide whether to measure “particle-like” or “wave-like” behavior after a photon has already passed through the double slits, and the measurement outcome seems to retroactively change the photon’s past path. The Copenhagen interpretation abandons a realist description of “the past,” while the information-update interpretation offers an alternative clear language. Unitary stage: computation without recording. From emission to arrival at the detection screen, the photon’s evolution is governed by the Schrödinger equation’s unitary dynamics. During this stage, the photon state is a path superposition, where the phase information includes all experimental apparatus settings (including later inserted measurement devices). This evolution is fully reversible: if all measurement devices are removed before the photon arrives, the system can cleanly return to its

initial state. Importantly, no information has been irreversibly written into the environment — it is like a computer program performing multi-path calculations in the background without committing the final result. Measurement as log submission. When the photon reaches the detection screen and is recorded (e.g., generating an observable current via the photoelectric effect), an irreversible information update occurs. The environment (detector atoms, electronic circuits, etc.) acquires definite eigenstate information. At that moment, the system must select one of the many computational paths as “factual history” and formally write it into the cosmic log. The key to the delayed-choice experiment is that the type of measurement apparatus (whether a which-path detector is placed, whether a beam splitter is inserted, etc.) determines which observable’s eigenvalue (position or momentum) is “submitted.” Because the unitary stage already contains all possible experimental configuration information, the final log entry is entirely self-consistent with the measurement setup, producing no logical contradiction. The illusion of “retrocausality.” Observers feel that “the past has been changed” because they are accustomed to thinking that history is fixed before measurement. In the information-update interpretation, before measurement (log submission), there is no definite classical history — only the correlation potentials described by unitary dynamics. Measurement does not discover a pre-existing fact but creates a new log entry. This entry is compatible with the subsequent classical causal chain but is independent of any “pre-measurement reality.” Thus, no need to invoke action-at-a-distance or backwards causation; it is sufficient to recognize that the moment of information update is the moment history solidifies. The delayed-choice experiment merely illustrates a nontrivial instance of this principle.

Creation and Annihilation of Virtual Particles: “Potentiality” vs. “Actualization” of Microscopic Time Updates

From an information perspective, the creation and annihilation of a virtual particle-antiparticle pair is a fully reversible quantum computation: the system starts from the vacuum, evolves into particle states, and then evolves back to the vacuum, ending in the same state. Because no energy or information leaks into the environment, the entire process leaves no permanent record in the cosmic log — it is merely a “virtual trial run” of the vacuum, contributing no passage of time. If, before annihilation, the virtual particles are perturbed by the environment (e.g., a black hole horizon, a strong electric field, or scattering with a detector), the particle information irreversibly diffuses into the environment and solidifies into real particles. This event creates a new entry in the global log, marking an irreversible solidification of a fact. Therefore, each “virtual-to-real” event constitutes a fundamental unit of microscopic time update in the universe. The statistical accumulation of many such events recovers the thermodynamic arrow of time, whereas fully annihilated virtual pairs are like cancelled computations and do not propel time forward.

Quantum Entanglement and “Faster-than-Light” Correlations: Holistic Collaborative Update, Not Signal Transmission

An entangled state is an indivisible quantum information whole. Measuring particle A triggers a global information update — because this update occurs within one time, the entire system’s state is atomically reconstructed. The state of particle B is not “notified”; it is automatically determined by the global update. This instantaneity is not signal transmission but the inherent cooperativity of the update: instantaneity = update cooperativity. No information is transmitted across spacelike intervals, so relativity is not violated.

2 Constructing an Alternate World

The author argues that the fundamental origin of the arrow of time is not quantum decoherence itself, but a basic asymmetry between disentanglement and entanglement: **The speed of quantum disentanglement is faster than light ($\gg c$), The speed of establishing quantum entanglement is limited by the speed of light ($\leq c$).** This asymmetry breaks time-reversal symmetry at a more fundamental level than decoherence, providing a physical basis for irreversibility that is more primitive than decoherence. Decoherence is merely the macroscopic emergent manifestation of this asymmetry when an open system couples to its environment. In the universe's quantum information processing, "disconnecting" is faster than "connecting." This asymmetry drives the unidirectional accumulation of log entries, thereby generating the passage of time that we perceive.

In a Penrose diagram, a spacelike region is depicted where particles move "faster than light." Now let us enter this alternate world and see what happens: how might a world whose physical rules are completely different from our own universe evolve into something bizarre and wondrous?

Analogous to the symmetric relationship between entanglement and disentanglement in our world, we define the Doctor Xin's rule for this fast universe:

- **The speed of establishing quantum entanglement is faster than light ($\gg c$);**
- **The speed of quantum disentanglement/decoherence is limited by the speed of light ($\leq c$).**

Since time arises from decoherence while our universe follows unitary evolution, causality is preserved despite superluminal interactions (no time reversal occurs). To comply with this rule, the elementary particles in this universe must have imaginary mass ($m^2 < 0$), and these particles always travel faster than light. Based on the above rules, we speculate on the properties of spacetime and matter in the fast universe: Properties of matter: Matter consists of tachyons with imaginary mass, with energy-momentum relation $E^2 = p^2 c^2 - |m^2| c^4$, and speed always exceeding the speed of light.

Properties of space: Space is not a pre-given smooth background but emerges from an entanglement network. The "distance" between two tachyons is determined by the strength of entanglement (stronger entanglement means shorter distance), rather than by coordinate differences. Space is essentially a discrete, arbitrarily connectable graph structure. Properties of time: Any signal travels far faster than light. There is no globally consistent temporal order. In the fast universe, there is no absolute "past" or "future"; only decoherence events provide local "before/after" relations that propagate at the speed of light. Time emerges from the decoherence process. This world appears as an infinitely boiling soup of entanglement: countless tachyons flash into existence instantaneously, then form a dynamic network through global entanglement. Entanglement in the network is slowly broken by local decoherence wavefronts ($\leq c$); after being broken, the tachyons become randomly entangled again, forming new connections. The overall picture is: fast generation of correlations (instantaneous) + slow erasure of correlations (at light speed), leading to incessant "bubble-like" fluctuations. There are no fixed particle trajectories, no eternal structures, only transient entanglement clusters that constantly appear and disappear. The core of the fast universe is: a globally coherent boiling soup + locally decoherent bubbles. "Emergence" is driven by the difference in timescales between entanglement and decoherence. Entanglement establishment is instantaneous (timescale ≈ 0), while decoherence propagation requires at least $\Delta t \geq \Delta x / c$. Thus, a cluster of tachyons first instantly "flashes" into an entangled structure, then after some time (depending on the size of the region), the decoherence wavefront sweeps through and breaks it apart. This is like bubbles continuously rising in a soup: rapidly generated, slowly dissipated. "Clustering" is not about spatial proximity, but about entanglement

proximity. In our world, “clustering” means being close in spatial position. But in the fast universe, even if two tachyons are billions of light-years apart, as long as their degree of entanglement is sufficiently high, they are considered “in the same bubble.” Thus, clustering is clustering on the entanglement network, not spatial co-location. Decoherence only operates within a “light-speed-accessible” range. This rule is crucial: for two tachyons to undergo decoherence (i.e., to break their mutual entanglement), they must satisfy $\Delta x \leq c \cdot \Delta t$ spatially. In other words, decoherence is a local, slow erasure mechanism. Consequently: Long-distance entanglements are never disturbed by decoherence $\rightarrow$ they form long-range stable correlations that constitute the “background structure” of the fast universe. Short-distance entanglements are constantly broken by decoherence and then re-established $\rightarrow$ they produce local dynamic fluctuations, i.e., those “boiling bubbles.” There is no global time in the entire world, only local temporal order provided by local decoherence events. Because there is no light-cone causal restriction, the whole universe cannot be described by a single time coordinate. Each decoherence event (propagating at the speed of light) defines a “before/after” within its range of influence, but there is no absolute order between different events.

... Wait a moment... this locally encapsulated decoherence region – is that not exactly our classical universe? Our ordinary world (the slow universe) is precisely a “decoherence bubble” within the fast universe.

When a decoherence event occurs in some region of the fast universe, the event propagates a decoherence wavefront outward at the speed of light. Inside the wavefront, global instantaneous entanglement is destroyed, and all interactions and signal propagation are limited to the speed of light c , thereby giving rise to the familiar physical rules of the slow universe (speed-of-light limit, causal order, locality, real-mass particles, etc.). This region inside the wavefront is a “decoherence bubble,” which is exactly our slow universe (our own universe). The bubble wall is the decoherence wavefront. Inside the bubble is an ordered, light-speed-limited local world; outside the bubble is the chaotic, instantaneously entangled boiling soup of the fast universe. Hence, the slow universe and the decoherence bubble in the fast universe are one and the same entity. We live inside the bubble and can only perceive physical laws below the speed of light.

3. Tachyon Sea Hypothesis of Cosmic Evolution

3.1. The Matrix: A Sea of Tachyons with Imaginary Mass

The ground state of the universe is a timeless, boundaryless quantum gravitational system whose fundamental degrees of freedom consist of a tachyon field with imaginary mass squared ($m^2 < 0$). This tachyon sea resides in a false vacuum state of extremely high energy density, obeys unitary evolution as a whole, and possesses global quantum entanglement. In this matrix, there exist no classical time or space dimensions; all physical degrees of freedom are in coherent superposition states.

3.2. Decoherence and the Birth of Our Universe

Due to quantum fluctuations, a spontaneous decoherence event occurs in some local region of the tachyon sea. Decoherence causes this region to condense from the false vacuum to the true vacuum, releasing vacuum energy. This process is accompanied by a topological phase transition: real-mass modes ($m^2 > 0$) emerge from degrees of freedom that originally only had imaginary mass, forming “tachyon-geometry” bound states (Planck-scale bubbles). The bubble wall expands outward at the speed of light. Inside the bubble, the decoherence process defines a local arrow of time (the radial direction of the bubble wall) and a three-dimensional space (the region enclosed by the bubble wall). Simultaneously, a speed-of-light limit and causality are established inside the bubble. A classical observable universe (a “slow universe”) forms inside the bubble.

3.3. Cosmic Expansion and Evolution

The continuous expansion of the bubble wall is equivalent to cosmic inflation and the subsequent accelerated expansion of space. The matter inside the bubble (real-mass particles, radiation) evolves according to the rules described by general relativity and quantum field theory: gravity dominates the formation of large-scale structures, while electromagnetic, strong, and weak forces govern microscopic processes. As expansion proceeds, the recession speeds between different matter regions may exceed the speed of light, but as long as they lie inside the same bubble, they remain causally connected. Cosmic evolution follows the second law of thermodynamics, irreversibly evolving from a low-entropy initial state to a high-entropy state.

3.4. Causal Isolation Leads to Particles Leaving Our Universe and

Dissolving into the Tachyon Sea

When the universe expands to a sufficiently large scale, the recession speed of some matter regions exceeds the speed of light, causing them to move beyond each other’s causal horizons. In the Big Freeze scenario, the horizon eventually engulfs everything, and every particle ends up at the center of its own horizon, forever losing contact with the outside. At that point, the quantum entanglement between these mosaic regions and the main body of the universe is severed, and they become causally isolated systems. Their superluminal behavior precisely corresponds to the superluminal motion in the tachyon sea. **Phase transition trigger:** For a causally isolated system containing only real-mass particles, its wavefunction undergoes a fundamental change. Within the framework of quantum gravity, such a system can evolve into a spacelike mode via tunneling or analytic continuation, i.e., its effective mass squared changes from positive to negative ($m^2 > 0 \rightarrow m^2 < 0$). The newly generated tachyons possess the same quantum properties as the parent tachyon sea and naturally become re-coherent. The information carried by the original particles is encoded into the global entanglement structure of the tachyon sea, achieving non-local storage of information. The entire evolution is unitary, information is conserved, and energy is preserved.

3.5. Cycle and Generation of New Bubbles

After receiving information and energy from causally isolated particles, the local fluctuation conditions in the tachyon sea are replenished. New decoherence events randomly and continuously occur in the matrix, forming new Planck bubbles. Each new bubble independently undergoes the complete process of birth, expansion, causal isolation, and return to the matrix. Different bubbles are causally separated and invisible to each other. Our observable universe is just one of many transient “slow-universe bubbles.” The bubble wall expands outward at the speed of light, so the tachyon sea outside the bubble can never be detected by observers inside the bubble.

4 Tachyon Dynamics Equations

In relativistic quantum mechanics, a tachyon with imaginary mass ($m^2 < 0$) is a relativistic object whose evolution is described by the Klein-Gordon (KG) equation:

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = \frac{m^2 c^2}{\hbar^2} \phi$$

When $m = i\mu$ is imaginary, the right-hand side becomes $-\frac{\mu^2 c^2}{\hbar^2} \phi$. This is itself the standard form of the tachyon field equation. The general solution of the imaginary-mass KG equation usually diverges exponentially ($e^{\pm \kappa t}$), corresponding to unstable tachyon condensation. To obtain a stable equation that is unitary, we need to impose closed-space boundary conditions. In a closed space (e.g., a box), boundary conditions quantize the wavenumber, allowing oscillatory solutions.

4.1 Dynamics Equations in Different Dimensions

One-dimensional case

Consider a one-dimensional tachyon of length L in the interval $[0, L]$ with Dirichlet boundary conditions $\phi(0, t) = \phi(L, t) = 0$.

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} = -\kappa^2 \phi, \quad \text{with } \kappa = \frac{\mu c}{\hbar} > 0$$

Separation of variables: let $\phi(x, t) = X(x) T(t)$.

Spatial part: $X_n(x) = \sin(k_n x)$, with $k_n = \frac{n\pi}{L}$, $n=1, 2, 3, \dots$

Time part: $\frac{1}{c^2} \ddot{T}_n = (k_n^2 - \kappa^2) T_n$.

Criterion: If $k_n > \kappa$, the right-hand side is positive, $\ddot{T}_n = +\omega_n^2 T_n$, solution $T_n \sim e^{\pm i\omega_n t}$ oscillatory in time, stable and unitary. If $k_n < \kappa$, the right-hand side is negative, solution $T_n \sim e^{\pm \kappa_n t}$ exponential divergence (unstable).

To keep tachyons stable, we cut off the low-frequency (long-wavelength) modes, ensuring that only modes with $k_n > \kappa$ exist in the system. This implies:

Physical realisation: the spatial length L must be smaller than some critical value.

$$L < L_{max} = \frac{\pi \hbar}{\mu c}$$

Alternatively, for a fixed length L , the imaginary mass μ must be sufficiently small to satisfy $\mu < \pi \hbar / (Lc)$, so that even the lowest mode $n=1$ satisfies the oscillation condition. In a one-dimensional closed space, the long-wavelength (low-frequency) instability of the imaginary-mass tachyon is cut off by the boundary conditions, leaving only short-wavelength, high-frequency stable oscillatory modes. This physical picture is analogous to a vibrating string in string theory under an imaginary-mass background.

Three-dimensional spherical symmetry

Dynamics in three dimensions and the Planck-scale issue. Stability criterion in the three-dimensional spherically symmetric case: consider an imaginary-mass field ϕ inside a sphere of radius R with homogeneous Dirichlet boundary condition $\phi(r=R, t)=0$. Write $\phi(\mathbf{r}, t) = \psi(r)T(t)$. The radial part uses spherical Bessel functions. To ensure the time part yields oscillatory behaviour: the radial equation for the spherically symmetric case (with angular momentum quantum number $l=0$ – the lowest mode, also the most dangerous because it is non-singular at $r=0$ and has the longest wavelength):

Radial equation:

$$\frac{d^2 u}{dr^2} + (k^2 + \kappa^2)u = 0$$

Where $u(r) = r R(r)$, $R(r)$ radial wave function k is the separation constant (real wavenumber). The equation becomes

$$\frac{d^2 u}{dr^2} + K^2 u = 0, \quad K^2 = k^2 + \kappa^2$$

The solution is $u(r) = A \sin(Kr)$ (finite at $r=0$).

Boundary condition $u(R)=0$ gives $KR=n\pi$, $n=1, 2, 3$ i.e.

$$k^2 + \kappa^2 = \left(\frac{n\pi}{R} \right)^2$$

The time part $\ddot{T} = -c^2 k^2 T$. The key is that k^2 must be positive for oscillatory solutions:

$$T(t) \sim e^{\pm i c |k| t}, \quad k^2 = \left(\frac{n\pi}{R} \right)^2 - \kappa^2 \quad \text{which means} \quad \left(\frac{n\pi}{R} \right)^2 > \kappa^2$$

The most dangerous is the lowest mode $n=1$, so stability requires $\pi/R > \kappa$

Thus the critical radius is $R < R_{max} = \pi/\kappa = \pi\hbar/(\mu c)$,

Does this approach the Planck scale? It depends on the value of the imaginary mass μ . The Planck

mass is $m_P = \sqrt{\frac{\hbar c}{G}} \approx 2.18 \times 10^{-8} \text{ kg}$, Planck length $l_P = \sqrt{\frac{\hbar c}{c^3}} \approx 1.62 \times 10^{-35} \text{ m}$. Set $\mu = \alpha m_P$

where α is dimensionless. Then $R_{max} = (\pi \times l_P) / \alpha$

- If $\alpha = \pi$ (i.e., $\mu \sim \pi m_P$), then $R_{max} = l_P$ – Planck scale.
- If $\alpha > \pi$ (imaginary mass larger than Planck mass), $R_{max} < l_P$, which is meaningless in current physics (quantum gravity dominates).
- If $\alpha < \pi$ (imaginary mass smaller than Planck mass), $R_{max} > l_P$ and can be much larger.

Therefore, the critical radius is not necessarily of the order of the Planck scale; it depends on μ .

If μ is small, macroscopically stable “imaginary-mass tachyons” can exist in large spaces.

General d-dimensional evolution equation and stability criterion

Considering that a sphere has the largest geometric “size”, in a d -dimensional space ($d \geq 3$) with a spherical cavity of radius R , the imaginary-mass field equation is

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla_d^2 \phi + \kappa^2 \phi = 0, \phi(r = R) = 0 \text{ with } \kappa = \frac{\mu c}{\hbar}$$

The stability criterion (unitary evolution condition) – that no mode diverges, i.e., all modes are oscillatory in time – is that the eigenvalue of the lowest angular momentum radial mode is greater than κ^2 . The condition is

$$\left(\frac{Z_d}{R} \right)^2 > \kappa^2$$

where Z_d , is the first zero of the d -dimensional radial Bessel function for zero angular momentum.

Critical radius formula:

$$R_{max}(d) = \frac{z_d}{\kappa} = \frac{z_d \hbar}{\mu c}$$

Variation of critical radius with dimension

Z_d is the first zero of the d -dimensional spherically symmetric Bessel function (zero angular momentum). Its numerical value increases monotonically with dimension d .

Dimension d	Radial equation type	First zero Z_d (approx.)	Critical radius R_{crit}
3	$j_0(z) = 0$	$\pi \approx 3.1416$	$3.14 \pi \hbar / (\mu c)$
4	$J_{1/2}(z) = 0$	≈ 3.8317	$\approx 3.83 \hbar / (\mu c)$
5	$j_1^{(5)}(z) = 0$	≈ 4.493	$\approx 4.49 \hbar / (\mu c)$
6	$J_{2/3}(z) = 0$	≈ 5.1356	$\approx 5.14 \hbar / (\mu c)$
d (even)	$J_{d-2/2}(z) = 0$	$\sim \frac{d}{2} + O(d^{-1/3})$	increases with d
d (odd)	spherical Bessel zero	$\sim \frac{d}{2} + a \left(\frac{d}{2}\right)^{1/3} + b + O(d^{-1/3})$	increases with d

Conclusion: $R_{\text{max}}(d)$ increases monotonically with d . Higher dimensions can accommodate larger stable imaginary-mass fields.

High-dimensional limit and physical implications

1. **No upper bound on dimension:** From a purely mathematical perspective, d can be any positive integer. For any d , as long as the radius $R < R_{\text{max}}(d)$, the imaginary-mass field inside the d -dimensional sphere is stable and oscillatory, and the evolution is unitary.
2. **Asymptotic behaviour for large d :** When $d \rightarrow \infty$, the first zero Z_d has the asymptotic form

$$z_d \sim \frac{d}{2} + \frac{1}{2^{2/3}} \alpha_1 \left(\frac{d}{2}\right)^{1/3} + \dots$$

where $\alpha \approx 2.338$ is the first zero of the Airy function. This means that in very high dimensions, the critical radius grows linearly with d : $R_{\text{max}} \sim \frac{d}{2} * \frac{\hbar}{\mu c} O\left(d^{\frac{1}{3}}\right)$

3. Physical picture:

- 3D: $R_{\text{max}} \approx 3.14/\kappa$
- 10D: $R_{\text{max}} \sim 10/\kappa$
- Extremely high dimensions: the stable region can be very large.

Explanation: the “hyperspherical Bessel functions” oscillate faster in higher dimensions, so the first zero occurs at a larger argument, allowing a larger cavity radius without triggering the exponential instability (radial energy is converted into angular energy).

The Planck-scale issue

If the imaginary mass is $\mu = \alpha m_P$, then

$$R_{\max}(d) = \frac{z_d}{\alpha} l_P$$

For fixed μ , $R_{\max}$ increases with d . Higher dimensions make it easier to avoid being compressed below the Planck scale. Only when $z_d \lesssim \alpha$, would $R_{\max} \lesssim l_P$ approach the Planck scale. For large d , this would require μ to be very large (a large α).

Conclusion above:

- No upper bound on dimension: d can be any positive integer, mathematically fully self-consistent.
- Higher dimensions are more “tolerant”: $R_{\max}(d)$ increases with d . Higher dimensions suppress the long-wavelength instability of the imaginary-mass field more strongly, allowing larger closed spaces.
- Unified evolution: in all dimensions, the field is expanded as a superposition of hyperspherical Bessel functions $j_l^{(d)}(kr)$ and time phases $e^{\pm i\omega t}$; as long as $R < R_{\max}(d)$, the evolution is unitary.
- Counter-intuitive discovery: for an imaginary-mass field, increasing the spatial dimension actually enlarges the scale of stable existence, rather than confining it to the Planck scale.

4.2 Perturbations

The spatial evolution of tachyons under small perturbations, and the shape selection effect: the stability of the system is determined by the “minimum characteristic scale”, i.e., oscillation along the “narrowest” direction. Consider a three-dimensional rectangular box with dimensions L_x, L_y, L_z with $L_x \ll L_y, L_z$. Stability criterion: the wavenumber must satisfy $k > \kappa$. The smallest wavenumber is determined by the smallest dimension $k_{\min} \approx \pi/L_x$ (because L_x is smallest). Dominant mode: to satisfy $k > \kappa$, the system must rely on the narrowest direction x . Therefore, the lowest-frequency vibration mode (the fundamental mode) indeed appears as a standing wave along the narrowest direction x , while being approximately constant (zero mode or very low-order mode) along the other directions y & z .

Physical picture: the instability (exponential divergence) of the imaginary-mass field corresponds to long-wavelength (low-frequency) perturbations. The narrowest dimension naturally cuts off the long wavelengths, like a pair of scissors trimming away the unstable “long waves”, forcing the system to oscillate at shorter wavelengths (higher frequencies).

Tachyon vibration forms: in the lowest-order approximation they appear as “strings”, but in the full picture they are “branes”.

1. **One dimension (string):** In one dimension (e.g., a line segment), the vibration is simply the standard string vibration: $\phi(x,t) \sim \sin(n\pi x) e^{\pm i\omega t}$. This is the “open string” picture.
2. **Two dimensions (membrane):** The situation becomes more complex. In a two-dimensional rectangular region, if $L_x \ll L_y$, the lowest mode is indeed $\phi(x,t) \sim \sin(\pi x/L_x) e^{\pm i\omega t}$. This looks like a

“string” vibrating in the xx direction. However, once higher modes are excited, mixed standing waves in both x and y directions appear (e.g., $\sin(\pi x/L_x) \sin(\pi y/L_y)$), which are membrane vibrations.

3. **Three dimensions and higher (supermembrane):** No longer simply a string. In a sphere or a rectangular box of dimensions $d \geq 3$:
 - Fundamental mode (lowest mode): due to symmetry or the minimum-size constraint, it may appear as a one-dimensional vibration in one direction (e.g., the spherically symmetric $l=0$ mode, which depends only on r).
 - Excited states: once the angular momentum $l > 0$, the vibration simultaneously extends over multiple dimensions, manifesting as complex multi-dimensional standing waves, described by hyperspherical Bessel functions and hyperspherical harmonics.

Mathematical essence:

Separation of variables:

$$\phi(\mathbf{r}, t) = R(r)Y(\Omega)e^{\pm i\omega t}$$

Time part $T(t) \sim e^{\pm i\omega t}$ always oscillatory – this is the “string-like” temporal behaviour.

Radial part $R(r)$: satisfies an equation of the form $d^2\mu/dr^2 + K^2\mu = 0$. This equation is mathematically identical in form to the one-dimensional string equation after a suitable transformation.

Angular part $Y(\Omega)$: this is the “extra decoration” brought by higher dimensions.

The “string vibration” essence is a mathematical manifestation of the radial equation. The field’s oscillation in the radial direction (from centre to boundary) obeys the same laws as a one-dimensional string, but the whole system is also overlaid with complex angular structures.

Considering the lowest energy levels and perturbative effects, the lowest-energy spatial configuration for stable tachyons should be a d -dimensional ellipsoid. The instability is determined by the smallest characteristic length (the radius $r_{\min}$ along the shortest axis). Along that shortest axis, the radial part behaves as a one-dimensional string vibration, and its energy exactly cancels the intrinsic tachyon energy. The angular part takes the form of multi-dimensional standing waves on a $(d-1)$ -dimensional membrane. In the case of an isolated space with no heat exchange, the interior “string” undergoes an eternal “pulsation” (contraction and expansion) in the radial direction. The surface “membrane” (with $l > 0$) undergoes eternal “breathing” or “pattern oscillations” in the angular directions.

4.3 Cutoff: Dimensional Matching and Imaginary Mass Matching

Dimensional matching:

To match the tachyon energy, space needs to possess a “self-organising” flexibility. The tachyon model eventually evolves into:

Dimension emergence: starting from chaotic initial conditions, the system spontaneously locks onto a specific dimension D_{opt} by adjusting (the parameters D and R).

Stable vibration: in this dimension, the imaginary mass of the tachyon is exactly balanced by the centrifugal barrier and geometric tension, and the system enters a PT-symmetric, rhythmically rotating vibrational state.

The system minimises the total energy by adjusting D to balance the “tachyon instability” against the “cost of the high-dimensional centrifugal barrier”. The effective potential V_{eff} roughly consists of two competing parts:

$$V_{eff}(D, R) \approx -\kappa^2(D) + \frac{C}{R^2} + \frac{D}{R^d}V$$

Tachyon term (negative energy): often scales with D (e.g., in string theory, $\kappa^2 \sim (D - D_{crit})$).

Geometry term (positive energy): includes the centrifugal barrier and spatial bending energy. Key point: higher dimensions (large D) have a strong centrifugal barrier, but “driving” so many dimensions itself requires energy (e.g., the tension needed to maintain a high-dimensional volume).

Dynamical process:

If D is too small, the tachyon term κ^2 dominates (large negative), and the system becomes extremely unstable (exponential blow-up).

If D is too large, the geometry term (e.g., D/R^2) explodes and the energy becomes too high.

There exists an optimal D_{opt} (e.g., 4, 8, 16, ...) such that the total energy V_{eff} takes a global minimum. In this dimension, the system neither explodes nor pays an excessively high geometric maintenance cost.

Imaginary-mass cutoff:

When the imaginary mass (instability) becomes too large, the system tends to “split” into multiple smaller stable units, rather than being compressed indefinitely. In essence, the instability of energy is resolved by a topological change of the spatial structure (the mechanism of tachyon condensation and spatial fragmentation). **Limit of a single cavity:** when κ (the absolute value of the imaginary mass) is extremely large, the radius R required for stability becomes extremely small ($R < \pi/\kappa$). However, R cannot be arbitrarily small – it is limited by the Planck scale (quantum gravity) or by diverging energy density. The total surface area of two small spheres is larger than that of a single large sphere, providing more “surface vibration modes” to dissipate or store energy, thereby reducing the energy density. “Very large imaginary mass” corresponds to a deep tachyon potential in physics. **Single-point instability:** at one point, the potential $V(\phi)$ is negative with a large absolute value, and the field strongly tends to roll toward some non-zero vacuum expectation value $\langle \phi \rangle = \pm v$. **Domain wall formation:** space spontaneously forms domain walls, with $\phi = +v$ on one side and ϕ

$= -\nu$ on the other. **Spatial partitioning:** these domain walls physically act as new spatial boundaries, dividing the original space into multiple regions.

Physical forbidden regions for the imaginary mass:

1. **Lower bound** – although mathematically μ can be arbitrarily small, physically this is meaningless. **Planck cutoff** ($\mu \gtrsim M_{Pl}$): Effective field theory breaks down when $\mu \ll M_{Pl}$. The characteristic scale of the system, $R \sim 1/\mu$, becomes much larger than the Planck length. In this case, the “flexible space” effectively degenerates into a large, flat classical background, quantum gravity effects (the flexibility of space itself) become negligible, and the tachyon instability is trivialised, losing the “self-organising” geometric effect.

Physical meaning: in this model, μ must be of the order of M_{Pl} or larger, so that space remains “compact and flexible” and the geometric tension can compete strongly with the tachyon instability.

2. **Upper bound** – fragmentation / non-locality cutoff ($\mu \lesssim M_s$):

Fragmentation mechanism: when μ is extremely large, the gradient of the tachyon potential $V(\phi) \sim e^{-T/\tau}$ becomes very steep, and the system no longer tends to maintain a single “vibrating ball”. Instead, it decays into multiple fragments (e.g., lower-dimensional D -branes) via tachyon condensation. **Non-locality:** in string field theory, when $|\mu^2|$ approaches the string scale M_s^2 , the theory exhibits strong non-locality (exponential operators), and point-particle field theory breaks down. The “survival window” for μ is $M_{Pl} \lesssim \mu \lesssim M_s$, where M_s is the string scale (usually $M_s \sim M_{Pl}$ or slightly smaller).

Joint cutoff of imaginary mass and optimal dimension:

Energy competition equation:

The effective potential competition is between

- Tachyon term (negative energy): $-\mu^2 \phi_0^2$ (drives instability)
- Geometry term (positive energy): $\sim D/R^2$ (high-dimensional centrifugal barrier + geometric tension)

Using natural units ($\mu = M_{Pl} = 1$). From the stability condition $R \sim Z_d/\mu$ and the asymptotic behaviour of the first zero of the d -dimensional Bessel function for large $Z_d \sim d/2$, the radius estimate is $R \sim d/2$ and the geometric energy estimate is

$$E_{\text{geom}} \sim \alpha_d / R^{d-2} \sim \alpha_d / (d/2)^{d-2}$$

Numerical search D_{opt} :

Because the geometric term E_{geom} grows extremely fast with d (e.g., factorial-like from the volume), the total energy $F(d)$ typically attains a minimum at a small integer d .

- If the growth is mild (e.g., linear), the optimum D_{opt} is often between 6 and 12.
- If the growth is strong (e.g., exponential, which is more reasonable), the optimum D_{opt} falls between 4 and 6.

This model is a classic “energy minimisation” problem, and the result is usually low single-digit dimensions like $d=4,5,6$.

Reference benchmark:

In the literature, the “tachyon density” is often compared to the Planck density $\rho_{P1} = M_{P1}^4$. In cosmological applications (e.g., tachyon dark matter studies), to prevent the tachyon density from dominating the early universe, one typically requires the potential energy $V(T) \ll M_{P1}^4$; otherwise, effective field theory fails. In an “isolated space”, the energy density is $\rho \sim \mu^4 \sim M_{P1}^4$.

Reasonable parameter range (natural units):

Parameter	Determining mechanism	Reasonable range
Tachyon imaginary mass μ	Planck cutoff (to avoid effective field theory breakdown)	$1 \lesssim \mu \lesssim 10$ (order of Planck mass M_{P1})
Emergent space dimension D	Energy minimisation (competition between geometric energy and tachyon energy)	4, 5, 6
Spatial radius R	Stability boundary (Planck-length order)	$R \sim D/2\mu \sim 1$

Logical chain: μ is large $\rightarrow R$ is small $\rightarrow$ high-dimensional geometric energy increases sharply $\rightarrow$ suppresses unbounded growth of $D \rightarrow$ the optimal D_{opt} is finite and small.

4.4 System Symmetry

The symmetry requirements of the “d-dimensional spherical cavity + imaginary-mass field + closed system” determine the “vibration patterns” of the field and the rules of evolution.

1. Core symmetries

The model possesses a hierarchy of symmetries from “global” to “local”:

Symmetry	Mathematical object	Physical meaning	Manifestation in the spherical cavity
Time translation	$t \rightarrow t + \Delta t$	Energy conservation (Hamiltonian H)	Frequencies ω are fixed; overall phase rotation of modes
Global U(1)	$\phi \rightarrow e^{-i\theta} \phi$	“Charge” conservation (if ϕ is complex scalar)	If the initial field is real, this symmetry is absent

Spatial rotation $SO(d)$	$r, \theta \rightarrow r, \theta'$	Angular momentum conservation	Most important: generates spherical harmonics Y_{lm}
Spatial reflection (P)	$\mathbf{r} \rightarrow -\mathbf{r}$	Parity	Splits L into even (positive parity) and odd (negative parity)
Time reversal (T)	$t \rightarrow -t$	Microscopic reversibility	Sends $\omega \rightarrow -\omega$; combined with P gives PT

2. $SO(d)$ rotational symmetry (high-dimensional angular momentum)

This is the most central symmetry in the model. It answers the questions “why do $\ell=0$ and $\ell>0$ modes exist?” and “how does the surface vibrate?”

Mathematical structure:

- Three dimensions ($d=3$): symmetry group $SO(3)$, generators L_i ($i=x,y,z$), quantum numbers ℓ, m .
- Higher dimensions ($d>3$): symmetry group $SO(d)$, more generators, quantum numbers $(\ell, m_1, m_2, \dots)$.

Physical consequences:

- **Mode classification:** the vibration modes of the field are strictly classified by the total angular momentum ℓ .
 - $\ell=0$ (scalar representation): spherically symmetric; the surface is “dead”; only radial string vibrations.
 - $\ell>0$ (tensor representation): the surface is “alive”; corresponds to vibrations on the $(d-1)$ -dimensional sphere S^{d-1} .
- **Degeneracy:** modes with $\ell>0$ are highly degenerate. For example, in 3D, one ℓ corresponds to $(2\ell+1)$ values of m . In higher dimensions, the degeneracy is even larger, meaning many different “surface patterns” share the same vibration frequency.

3. How do symmetries constrain “vibration”?

- **Radial direction is a “string” – an $SO(d)$ singlet:** due to $SO(d)$ symmetry, the radial direction r is the only “special direction”. It must be independent of angles. Therefore, the $\ell=0$ fundamental mode is necessarily a purely radial vibration – this corresponds to a “one-dimensional string”.
- **Angular directions are a “membrane” – a non-trivial $SO(d)$ representation:** modes with $\ell>0$ have their dynamics in the angular directions. Mathematically, they correspond to eigenfunctions of the Laplace-Beltrami operator on the sphere S^{d-1} . These are no longer “string” vibrations but surface waves on the $(d-1)$ -dimensional hypersphere.

Symmetry forces the decomposition of vibrations into “1D (radial string) $\times$ (d-1) (angular membrane)”.

4. The special role of PT symmetry

- In a model with imaginary mass ($m^2 < 0$) and closed boundaries, PT symmetry is not merely a “nice extra” but a “lifeline” that maintains the physical viability of the system. In the classical universe, the usual case ($m^2 > 0$): H is Hermitian, eigenvalues $\omega^2 > 0$, the system is automatically stable.
- In the tachyon sea ($m^2 < 0$): H is not Hermitian; eigenvalues ω^2 can become negative (ω imaginary, leading to exponential divergence).
- **Salvation by PT:** the closed boundary conditions select discrete wavenumbers k such that when $k > |\kappa|$ (the stability condition $\omega^2 = k^2 - \kappa^2 > 0$), although H is non-Hermitian, PT symmetry is not spontaneously broken, guaranteeing that ω is real and evolution is unitary.

5. Influence of dimension D

In higher dimensions, the symmetry is correspondingly enhanced:

- D=3: SO(3) symmetry, angular momentum ℓ is an integer.
- D=4: SO(4) $\approx$ SU(2) $\times$ SU(2); the symmetry is higher, and mode classification is finer.
- D $\rightarrow\infty$: extremely high symmetry; the system approaches mean-field behaviour.

6. Geometry-matter duality

In the tachyon sea model, because space adapts to the quantum dynamics of tachyons, the vibration modes of the tachyon are identical to the vibration modes of the enveloping space. The “particles” and “space” that observed are not independent entities.

- **Internal perspective (string theory / particle physics):** one sees high-energy, imaginary-mass “tachyons” vibrating.
- **External perspective (loop quantum gravity / geometry):** one sees the violent vibration of the geometric envelope that contains them.

The two are equivalent: they are two descriptions of the same physical reality, just like the “shape” and the “vibration modes” of an object.

A stable tachyon picture is as follows: a small sphere with a diameter close to the Planck scale, an imaginary mass between 1 and 10 Planck masses, and vibrating inside a compact, nearly ellipsoidal space of 4–6 dimensions. Symmetry forces the decomposition of spatial vibrations into “1D (radial string) $\times$ (d-1)D (angular membrane)”, described by hyperspherical Bessel functions and hyperspherical harmonics. This embodies a duality under the physical picture: the internal perspective is that of a high-energy tachyon with imaginary mass vibrating as a string (string theory); the external perspective is that of a geometric envelope with high curvature, negative energy, and violent oscillation (loop quantum gravity). This geometric envelope is the excited graviton. Since the total energy $E_{total}=0$, the graviton and the tachyon are tightly coupled inside space. Externally, only changes in geometry and

volume are manifested; these vibrational information are encoded on the spatial surface. The graviton observed from outside the space does not emit energy, but only transmits information, and the evolution is unitary.

5. Conversion between Imaginary and Real Mass

5.1 Tachyon condensation from imaginary mass to real mass (from the tachyon sea to our universe)

Sen A. *et al.*, in their studies of tachyon condensation [6–10], discovered that an unstable *D*-brane system undergoes a series of fundamental physical changes through the evolution of the tachyon field: **Conversion from imaginary mass to real mass**. This process can be intuitively understood as the system sliding from an unstable state to a stable one. *Initial state (imaginary mass)*: the system sits at the peak of an energy mountain. Any tiny perturbation makes it fall; this is the inherent instability. *Final state (real mass)*: when the system slides down to the bottom of an energy valley, it becomes stable. In this new state, any small perturbation does not cause the system to collapse; moreover, the perturbations themselves can be quantized into particles with normal, positive mass squared – i.e., they acquire “real mass”. **Energy release and the emergence of a stable vacuum**. In the unstable initial state, the *D*-brane/anti-*D*-brane system contains enormous energy (the tension of the branes). When tachyon condensation occurs, the system releases this energy. *Energy release*: as the tachyon field moves toward the bottom of the energy valley, the negative energy it generates exactly cancels the positive energy inherent in the *D*-brane and anti-*D*-brane. Ultimately, the total energy of the entire system becomes zero. *Supersymmetry restoration*: total energy returning to zero means the system returns to the “cleanest”, most fundamental vacuum state. A key feature of this state is the restoration of a higher-level spacetime symmetry called “supersymmetry”. **Dimensional reduction of space**. Tachyon condensation can not only change the energy state but also, through spatial inhomogeneity during condensation, create lower-dimensional objects. *Dimensional reduction*: on a two-dimensional plane, a structure with energy concentrated on a one-dimensional line is formed; this line-like structure is a lower-dimensional object. *Generation of lower-dimensional D-branes*: similarly, in string theory, a spatially inhomogeneous tachyon condensation configuration (e.g., a kink or vortex) naturally generates a stable structure whose energy is highly concentrated on a lower-dimensional subspace.

5.2 Inverse tachyon condensation from real mass to imaginary mass (from our universe to the tachyon sea)

Inverse tachyon condensation describes the complete process by which real-mass matter, under extreme gravitational conditions, undergoes spatial compression, causal disconnection, and energy reversal, eventually transforming into an imaginary-mass tachyon mode and returning to the vacuum ground state [6–16]. The physical origin of this mechanism can be traced to the following key steps: **1 Causal disconnection and gravitational spatial compression**. When a real-mass particle (or a clump of matter) is gravitationally compressed to the scale of its Compton wavelength or even the Planck scale, its local energy density rises sharply, and the spacetime curvature reaches extremes. Under such extreme conditions, the causal link between the particle and the external universe is effectively severed – it can no longer exchange information with the outside world via subluminal signals, and its own quantum state becomes the sole dominant physical degree of freedom in that

region. Causal disconnection provides the prerequisite for all subsequent non-trivial quantum gravitational effects. 2 **Energy reversal and the flipping from real to imaginary mass.** After causal isolation, the effective field theory parameters describing the particle are fundamentally reshaped. In the low-energy, weak-field limit, the effective potential of a scalar field has a minimum at $\phi=0$, corresponding to positive mass squared and stable real-mass excitations. However, when the energy density or curvature scalar exceeds a critical threshold, gravitational corrections dominate the behaviour of the effective potential, causing the second derivative at $\phi=0$ to turn from positive to negative; the original stable vacuum no longer exists. The particle is no longer a real-mass excitation with positive mass squared, but instead becomes a localised unstable mode – this is the core step of inverse tachyon condensation: the sign flip of the mass squared directly causes the material degree of freedom to transition from the timelike branch ($m^2>0$) to the spacelike branch ($m^2<0$). 3 **Tachyon emission and dimensional increase.** After the real-mass particle transforms into a tachyon mode, the process here is the inverse of the tachyon condensation described by Ashoke Sen: the *generation* (rather than disappearance) of tachyon modes marks the transition of the system from a stable vacuum to an unstable configuration. In this inverse process, the tachyon field rolls down from the maximum at $\phi=0$, and its dynamics naturally induce a change in spatial topology. Crucially, a spatially inhomogeneous tachyon configuration can yield topological soliton solutions – an unstable tachyon mode originally localised on a $(d-1)$ -brane can, through configurations such as kinks or vortices, manifest as a stable lower-dimensional D-brane in a higher-dimensional world of a D-brane. In other words, the dimensional increase from a $(d-1)$ -brane to a D-brane does not come from nowhere; it is a dimensional transition driven by the annihilation-creation mechanism of brane-anti-brane pairs during inverse tachyon condensation. This picture forms a mirror image of the logic described in Sen’s classic 1998 paper – “D-brane-anti-D-brane systems can produce lower-dimensional D-branes via tachyon condensation” – in the sense that in the inverse process, real-mass matter on a lower-dimensional brane, via tachyonisation, returns to the geometric degrees of freedom of a higher-dimensional brane. 4 **Return to the vacuum and information conservation.** After losing the constraint of a local light-cone structure, the tachyon mode can propagate in branches of quantum gravity that lack classical time, and re-cohere with the background tachyon sea of the vacuum. This process is not a loss of information, but rather the conversion of quantum information from local, classical encoding into non-local, quantum entanglement encoding. The unitarity of the total wavefunction guarantees information conservation, while the vacuum ground state – as the mother sea of all excitation modes – naturally receives the returning matter degrees of freedom. This picture is highly consistent with the analytic continuation of the timeless cosmological wavefunction between different branches in the Wheeler-DeWitt equation, and also provides a potential background explanation for the cosmological constant problem: the vacuum energy of the mother sea ground state is defined as zero, and the vacuum energy of our observable universe is merely a residual excitation relative to this ground state.

5.3 Evolution Mechanism of Imaginary and Real Mass Based on High-Dimensional Spherical Geometry

This section aims to construct a self-consistent dynamical evolution mechanism describing the mutual transformation between imaginary mass (tachyons) and real mass within a high-dimensional spherical (D-brane) model. By introducing the eigenvalues of the spherical Laplacian as boundary

constraints, a mass evolution formula isomorphic to the logic of tachyon condensation in string theory is derived, revealing the intrinsic connection between dimensional collapse and the imaginary/real mass transition. In the existing string theory framework, the instability of a D_p -brane originates from the tachyon mode in its open string spectrum, manifesting as an imaginary mass state with $M^2 < 0$. Through deformations of boundary conformal field theory (BCFT), the literature demonstrates that tachyon condensation can lead to a reduction in dimension (e.g., $D25 \rightarrow D24$) and, in the new vacuum, eliminate the imaginary mass to produce real mass excitations. This section formalizes this process from a purely geometric-dynamical perspective into an independent high-dimensional spherical evolution model. Consider a sphere S^d of initial dimension d and radius R as the system's foundation. On this sphere, define a tachyon field $T(t, \Omega)$, where Ω denotes the angular coordinates. The system initially sits at a local maximum of the tachyon potential, with an intrinsic mass squared negative: $M_{intrinsic}^2 = -\mu^2$, ($\mu^2 > 0$), corresponding to the imaginary mass phase. To incorporate dimensional effects, expand the field TT in spherical harmonics $Y_\ell(\Omega)$:

$$T(t, \Omega) = \sum_{\ell} T_{\ell}(t) Y_{\ell}(\Omega)$$

where ℓ is the angular momentum quantum number. The spherical harmonics satisfy the Laplace equation on S^d :

$$\nabla_{S^d}^2 Y_{\ell}(\Omega) = -\frac{\ell(\ell + d - 1)}{R^2} Y_{\ell}(\Omega)$$

Here $\frac{\ell(\ell + d - 1)}{R^2}$ is the positive definite energy term introduced by spherical geometry, central to the subsequent mass evolution. Within an effective field theory framework, for each mode $T_{\ell}(t)$ we construct the Lagrangian:

$$L = \frac{1}{2} \dot{T}_{\ell}^2 - \frac{1}{2} \frac{\ell(\ell + d - 1)}{R^2} T_{\ell}^2 - V(T_{\ell})$$

where $V(T_{\ell})$ includes the $-\mu^2$ tachyon potential. Varying T_{ℓ} yields the equation of motion:

$$\ddot{T}_{\ell} - \frac{\ell(\ell + d - 1)}{R^2} T_{\ell} + V'(T_{\ell}) = 0$$

Reorganizing into a standard wave equation, we define the effective mass squared M_{eff}^2 via:

$$\ddot{T}_{\ell} = \left[\frac{\ell(\ell + d - 1)}{R^2} - V''(T_{\ell}) \right] T_{\ell}$$

At the initial moment ($T_{\ell} \approx 0$), $V''(0) = \mu^2$, giving the core formula for imaginary/real mass evolution:

$$M_{eff}^2(R, \ell) = -\mu^2 + \frac{\ell(\ell + d - 1)}{R^2}$$

This formula clearly shows how the nature of the mass evolves with geometric parameters. In the imaginary mass regime, when the sphere radius R is small (Planck-scale) or the angular momentum ℓ is low such that $\frac{\ell(\ell + d - 1)}{R^2} < \mu^2$, we have $M_{eff}^2 < 0$ – the system exhibits imaginary mass tachyons, corresponding to an unstable high-dimensional D-brane. At the critical transition point:

$R = R_c = \sqrt{\frac{\ell(\ell + d - 1)}{\mu^2}}$, $M_{eff}^2 = 0$, crossing the boundary between imaginary and real mass. In the real

mass regime, when R continues to increase (spatial expansion) or ℓ grows so that the geometric term dominates, $M_{eff}^2 > 0$ – the imaginary mass disappears and the system evolves into stable real-mass particles. This derivation also reveals the dynamical picture of dimensional reduction: the initial

high-dimensional sphere S^d constrains the tachyon via boundary conditions, rendering it imaginary mass; as R evolves or tachyon condensation occurs, the system rolls to a new vacuum. In this new stable state, the perturbative excitation spectrum no longer contains tachyons but exhibits real mass, and the effective dimension may collapse to a lower one (e.g., the three dimensions of interest). In conclusion, by incorporating the geometric constraints of a high-dimensional sphere S^d , this section successfully derives dynamical equations for the mutual transformation between imaginary and real mass. The result shows that imaginary mass is not an exotic exception but a natural consequence of specific geometric conditions (small radius R or low mode ℓ). By adjusting the geometric parameter R , the system smoothly undergoes a phase transition from imaginary mass (unstable high dimension) to real mass (stable low dimension), providing a clear and self-consistent mathematical physics framework for understanding tachyon condensation and the origin of dimensions.

5.4 Critical Temperature of the Imaginary-Real Mass Phase Transition in High-Dimensional Spherical Geometry

This section aims to derive, based on a high-dimensional spherical collapse model, the critical temperature of the phase transition between imaginary and real mass during the evolution of the universe from 4 to 3 dimensions, and to demonstrate the equivalence of this temperature to the characteristic temperature of the cosmic inflation epoch. The entire derivation is based on the principle of Helmholtz free energy minimization in finite-temperature quantum field theory. The system state is described by its radius R , and the equilibrium state is determined by the vanishing first variation of the free energy $F(R, T)$: $\frac{\partial F(R, T)}{\partial R} = 0$. The free energy $F(R, T)$ consists of three physical contributions. The first term is the geometric stiffness term A/R^2 , originating from the high-dimensional spherical geometry constraint (Section 5.3 of the reference). Its physical meaning follows directly from the eigenvalues of the Laplacian on a high-dimensional sphere. The eigenvalue $\frac{\ell(\ell+d-1)}{R^2}$ of the radial equation is obtained via separation of variables. To maintain dimensional consistency and simplify the expression, this part is combined into the coefficient A . This term represents the cost of spatial curvature: when the radius R is very small, the space is highly curved and the energy rises sharply, acting like an elastic potential energy resisting compression. The second term is the thermal fluctuation term BT^2R^2 , originating from the high-temperature expansion of quantum field theory (Matsubara formalism). Its physical meaning is to introduce the temperature T . In finite-temperature QFT, the free energy density in the high-temperature limit generally contains a term proportional to T^4 . Considering that the system volume is proportional to R^d (where d is the dimension), after renormalization or variable substitution, its contribution to the free energy typically takes the form BT^2R^2 (especially near $d=4$). This term represents the expanding force of thermal noise: the higher the temperature T , the more intense the thermal motion, and the larger this term, tending to "inflate" the space. The third term is the imaginary mass driving term $-\mu^2R^2$, originating from the instability of the tachyon sea or high-dimensional state (Sections 3.1 and 5.3 of the reference). Its physical meaning corresponds to the intrinsic imaginary mass squared $-\mu^2$ mentioned in the reference, which drives the system away from the origin towards larger radii. This term represents the inherent expansion tendency of high-dimensional space and is the root cause of system instability before dimensional collapse. Summing the three contributions, the total free energy of the system is:

$$F(R, T) = \frac{A}{R^2} + BT^2 R^2 - \mu^2 R^2$$

Imposing the equilibrium condition:

$$\frac{\partial F}{\partial R} = -\frac{2A}{R^3} + 2(BT^2 - \mu^2)R = 0$$

Solving for the temperature-dependent radius gives:

$$R(T)^2 = \frac{A}{BT^2 - \mu^2}$$

This equation clearly demonstrates the jump mechanism. In the high-temperature stable regime ($T > T_c$), when the temperature is extremely high (e.g., the Planck temperature T_P in the early universe), the thermal fluctuation term BT^2 dominates over the imaginary mass term μ^2 , making the denominator positive. The system then possesses a finite, stable small-radius solution, corresponding to the "tachyon universe" (high-dimensional, imaginary mass, superluminal) described in the reference. At the critical point ($T = T_c$), the critical temperature is defined by $T_c = \mu/\sqrt{B}$. Then $BT_c^2 - \mu^2 = 0$, leading to $R(T) \rightarrow \infty$. This means that at T_c , the original small-radius vacuum state becomes completely unstable. In the low-temperature regime ($T < T_c$), the denominator becomes negative, so the real solution for R disappears. The system can no longer maintain equilibrium by smoothly adjusting R ; instead, it must undergo a first-order phase transition via quantum tunneling, jumping from an unstable "small universe" to a completely new, macroscopic "large universe". This corresponds to the "slow universe" (three-dimensional, real mass, speed-of-light limit) in the reference. Thus, the critical temperature derived from finite-temperature free energy minimization is $T_c = \mu/\sqrt{B}$, and under Planck-scale boundary conditions this coincides with the Planck temperature, precisely matching the epoch of cosmic inflation. After inflation ends, the radius R freezes and a thermodynamic barrier isolates dimensions, preventing further reactions. Thus, while 4D objects may pass through our universe, the frozen geometry keeps them stable and non-interacting with our 3D space.

Reviewing the critical geometric condition in the tachyon sea model, the effective mass squared M_{eff}^2 of the system is determined by the geometric constraint of a 4-dimensional sphere and the intrinsic imaginary mass:

$$M_{\text{eff}}^2(R, l) = -\mu^2 + \frac{l(l+d-1)}{R^2}$$

At the boundary between imaginary and real mass, the radius reaches its critical value: $R_c = \frac{\sqrt{l(l+3)}}{\mu}$

According to cosmological principles, the energy density ρ is inversely proportional to the fourth power of the curvature radius R . At the critical point, the energy density ρ_{phase} can be expressed as:

$$\rho_{\text{phase}} \propto \frac{1}{R_c^4} = \frac{\mu^4}{l^2(l+3)^2}$$

Adopting the Planck-scale boundary condition you have set, take $\mu \sim M_P$ (the Planck mass) and the angular momentum mode ℓ on the order of Planck scale ($\ell \sim 1$), the above expression simplifies to: $\rho_{\text{phase}} \approx M_P^4$. This indicates that the energy density at the phase transition is approximately the Planck density. In the hot big bang cosmological model, the energy density in the radiation-dominated era is related to the temperature T by $\rho \propto T^4$. Substituting the critical density yields the phase transition critical temperature T_c : $T_c^4 \approx M_P^4$, Hence: $T_c \approx M_P$. That is, the critical temperature

is approximately the Planck temperature $T_P \approx 1.4 \times 10^{32} \text{K}$. This derived critical temperature corresponds exactly to the temperature of the cosmic inflation epoch. The physical mechanism is as follows: in inflationary theory, the universe around the Planck time ($t \sim 10^{-43} \text{s}$) is at an extremely high temperature, near the Planck scale, and may have been in a high-dimensional symmetric state (such as the 4-dimensional sphere in the tachyon sea model). When the universe expands and cools to $T_c \approx M_P$, the radius R of the 4-dimensional sphere precisely expands to the critical value R_c . At this moment, the geometric constraint term $\frac{l(l+3)}{R^2}$ balances the imaginary mass term μ^2 , triggering dimensional collapse (from 4 to 3 dimensions) and the imaginary-real mass transition. This derivation shows that the "dimensional reduction" phase transition in the model occurs, on the cosmological timeline, exactly at the middle-to-late stage of inflation. In summary, the phase transition critical temperature $T_c \approx M_P$ derived from geometric dynamics perfectly matches the temperature of the cosmic inflation epoch in standard cosmology, providing strong quantitative support for the hypothesis that "dimensional origin and inflation share a common source."

6. Energy Equation (TOE)

In the "Tachyon Sea" model, space originates from the rigidity requirement for stable existence of discrete quantum excitations (the "tachyons") that possess imaginary mass. For each "tachyon" to exist as a physical entity rather than a mathematical singularity, it must intrinsically carry a "Planck-scale" geometric envelope. More generally, the first principle of the universe under the tachyon sea hypothesis is: **All stable energy must be confined by its own geometric envelope, and the total energy of the two is zero.**

$$\langle \hat{H}_g \rangle + \langle \hat{H}_m \rangle = 0$$

where $\langle \hat{H}_g \rangle$ is the average expectation energy of the geometric (gravitational) part, and $\langle \hat{H}_m \rangle$ is the average expectation energy of the matter part (including fields, particles, dark matter, etc.). The expectation value of the total energy (gravitational energy + matter energy) of the universe is zero. The microscopic equation is the Wheeler-DeWitt equation: $\hat{H}|\psi\rangle = 0$

Space is flexible, plastic, and self-adaptively curved in response to internal matter-energy. There is no space without matter (a negatively curved surface would directly collapse); nor is there matter without space (a naked singularity would lead to infinite density). The origin of space is neither a pre-existing empty stage nor an abstract, entropic mathematical illusion. Matter and space are a natural physically indivisible (inseparable whole). The macroscopic continuous spacetime we perceive is precisely the emergent condensate formed by the topological phase transition, mutual fusion, and freezing of countless "microscopic geometric envelopes" at low energies. In short, space is the "coat" that matter has to "wear" in order to exist – a continuous illusion emerging at macroscopic scales from a microscopic discrete geometric structure. The space of our universe appears large, flat, and hard to disturb because all the matter in our universe jointly and stably maintains a region far larger than the visible universe. The space of our universe is a nearly uniform sphere; the boundary of the universe has negative curvature and negative energy, while black holes, regions that have lost causal contact with horizons, and the projections of tachyons passing through our universe have positive curvature and positive energy.

Absolute space acts as a "regulator" that adjusts the internal matter-energy. For an open space, the curvature of our universe cooperates with the matter distribution in the most "energy-efficient"

way. This is the principle of least action in general relativity: given a matter distribution, the choice of spacetime metric makes the total action take a minimum value. In other words, the way spacetime curves is precisely the twisting that yields the “lowest energy”. It can also be understood as the local space “compromising” to respond to distant space so as to satisfy the condition of total energy zero. The equation reduces to the classical Einstein field equations of general relativity. The total action is $S = S_{\text{Grav}} + S_{\text{Matter}}$

The gravitational part (Einstein-Hilbert action): $S_{\text{Grav}} = \frac{c^4}{16\pi G} \int R \sqrt{-g} d^4x$ where R is the Ricci scalar (the scalar curvature of spacetime) and $\sqrt{-g} d^4x$ is the spacetime volume element. The matter part: $S_{\text{Matter}} = \int L_{\text{Matter}} \sqrt{-g} d^4x$, where L_{Matter} is the Lagrangian density of the matter. Applying the principle of least action, we set the variation of the action with respect to the metric $g_{\mu\nu}$ to zero, $\delta S = 0$.

This gives

$$\frac{\delta S_{\text{Grav}}}{\delta g^{\mu\nu}} + \frac{\delta S_{\text{Matter}}}{\delta g^{\mu\nu}} = 0$$

The first term yields $\frac{c^4}{16\pi G} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right)$, and the second term is defined as $-\frac{1}{2} T_{\mu\nu}$.

Thus,

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

7. Physical Philosophy Framework under the Tachyon Sea Hypothesis

7.1 Decoherence Cutoff

If space is intrinsic to matter, then when the decoherence bubble (our universe) expands, the newly created real particles also bring their own space with them. This process has no inherent stopping mechanism (it does not require the consumption of external resources). The cosmic bubble would expand indefinitely, eventually swallowing the entire tachyon sea. This directly contradicts the assumption that the “tachyon sea is an eternal background”. To prevent an endless, eternal decoherence evolution, the cosmic evolution based on the tachyon sea model requires a **decoherence cutoff mechanism**: a flat, empty space maintains a total energy of zero, but under the perturbation of vacuum energy (high-dimensional tachyons passing through our universe), it cannot remain flat and uncurved. The curved space couples with the high-dimensional tachyon space, and eventually this spacetime region is torn apart. **Cosmic causality (causal relations)**: causality depends on the observer’s position. The isolated islands that have lost causality are of two types: 1 **Causal islands**: the region enclosed by each observer’s own observable universe boundary (the particle horizon). This boundary is relative and depends on the observer’s position. Different observers have different causal islands, which can overlap. Two mutually unconnected causal islands are bidirectionally isolated. 2 **Spacelike islands**: there exists an absolute, observer-independent spacelike separation between two (or more) events, meaning no causal connection is possible between them. This is not a spatially continuous region but a relational state – an isolated pair or set of events. A black hole is a one-way isolated spacelike island. In this model, within a causal island, the eternal decoherence process must be halted by **spatial tearing**. When the

total matter energy approaches the Planck mass, the local space undergoes a “big rip” rather than a “big freeze”. The curling-up process of the torn space is precisely the **inverse tachyon condensation** process.

7.2 Vacuum Energy

The energy carried by a single tachyon when it passes through the classical universe. Using the Planck frequency as a cutoff, the estimated vacuum energy density is about $\sim 10^{113} \text{ J/m}^3$, which is of the same order as the Planck energy density from inverse tachyon condensation, $\rho_P = \frac{c^7}{hG^2} \approx 4.6 \times 10^{113} \text{ J/m}^3$. If the vacuum really possessed such a Planck-scale energy density, the universe would immediately be gravitationally bent into an extremely tiny bubble and would never expand as smoothly as it does today. In the tachyon sea cosmic model, vacuum energy is not an intrinsic mysterious energy of “empty space”, but rather the geometric disturbance caused by high-dimensional tachyons passing through the bubble wall and acting on our universe. This energy disturbance manifests only as a Planck-scale geometric energy of space. It is the “ready but not yet released” energy of high-energy, high-dimensional tachyons.

7.3 Black Holes

When matter in our universe is compressed to the extreme, it transforms into tachyons with imaginary mass. These tachyons exist in higher dimensions and move faster than light; they are wrapped in self-adaptive spatial bubbles and quickly escape the black hole, merging into the tachyon sea. From the perspective of the combined system of the tachyon sea and our universe, the total energy is conserved. The curvature of space around a black hole – the black hole continuously “picks up” matter, creating a “debt” of space borrowed from our universe, resulting in a lack of space and a positive-energy curvature of our world. From the viewpoint of matter in our universe alone, the matter is lost, but from the combined perspective of space plus matter energy in our universe, the total energy remains unchanged. The information swallowed by the black hole is encoded in the spatial bubbles of the tachyons and dissolves into the tachyon sea. The whole process simply returns information from our universe back to the tachyon sea. **The role of black holes in the “tachyon sea” model: the “phase-transition recycling station” of the universe.** In the tachyon sea model, the nature of a black hole is not simply a gravitational trap, but a critical device that triggers a dimensional phase transition. It connects the “real-matter universe” with the “imaginary-mass tachyon sea” and is a key link in the cosmic matter cycle. **Core mechanism: phase transition from real mass to imaginary mass. Traditional black hole paradox:** In general relativity, matter falling into a black hole’s singularity seems to lose information forever, violating quantum mechanical unitarity. *Solution in the tachyon model:* Deep inside the black hole (close to, but not necessarily reaching, the singularity), when matter is compressed to Planck-scale energy densities, a fundamental topological phase transition occurs: **Space “melts”:** The 3+1 dimensional continuous spacetime structure of our universe disintegrates, returning to the high-dimensional, discrete “tachyon sea” background. **Matter “reduces”:** Real-mass particles (e.g., protons, electrons) that were originally “frozen” in three-dimensional space lose their spatial confinement, their quantum states decohere, and they transform into imaginary-mass excitations in the high-dimensional tachyon sea.

Resolution of the black hole information paradox: The mainstream holographic principle holds that information is encoded on the two-dimensional event horizon of a black hole, leading to the logical difficulty of “how does information climb from the interior to the surface?” The tachyon model offers an explanation: information is not lost. During the “real $\rightarrow$ imaginary” phase transition, the quantum information of the infalling matter is converted from the encoding mode of “particle states” into the encoding mode of the topological structure of the high-dimensional tachyon sea. The information is encoded **at the phase transition boundary**, not on the event horizon. Due to the replicability of information, the true “information screen” is stored both on the classical geometry of the event horizon (the surface of the black hole) and in the dynamical process of the “phase transition interface” between the real universe and the tachyon sea. This interface is physical, not purely geometric.

The cycle of black holes and the universe: the “breathing” of an open system: The tachyon sea model views the universe as an open system embedded in the background of the “tachyon sea”. The tachyon sea, via low-energy phase transitions (such as inflation), “freezes out” space and real matter, creating and nourishing our universe. Black holes are the key **return channels**: *Internal consumption and dilution*: The universe expands, and matter effectively increases in entropy due to decoherence. *Return (recycling)*: Black holes, through high-energy phase transitions, take the used, highly ordered real matter (e.g., stars) and recycle it, degrading it and returning its information and energy in the form of imaginary mass to the tachyon sea background, providing raw materials for possible new creation. Thus, black holes are not endpoints but converters of matter forms. They resolve the information paradox and provide a dynamic, non-equilibrium picture of cosmic evolution, avoiding the ultimate fate of “heat death”.

7.4 Dark Matter

Because tachyons are dimensionally isolated and of Planck size, they do not participate in the electromagnetic interactions of our universe. However, their geometric projection in our universe possesses sufficiently high mass, making them ideal candidates for perfect dark matter. Their dynamical evolution from the Big Bang to the present is as follows:

Grand inflation:

Higher-dimensional tachyon-driven inflation model: The inflationary period of the universe can be interpreted as the interaction phase between the high-dimensional tachyon field and the three-dimensional brane universe. The driving force of inflation originates from the potential energy of the high-dimensional tachyon field, which is projected into three-dimensional space via brane coupling, converting into an effective vacuum energy density. The trigger of inflation corresponds to the three-dimensional brane universe moving into a high-density region of the tachyon field in the extra dimensions, where strong tachyon condensation occurs. The superluminal expansion, which appears acausal to a three-dimensional observer, is due to the fact that perturbations of the tachyon field establish causal connections in the extra dimensions (higher-dimensional causality). This model naturally solves the initial conditions problem of inflation: the uniform distribution of the high-dimensional tachyon field guarantees the initial uniformity of three-dimensional space, while the coupling between the brane and the tachyon field automatically provides the negative pressure needed to drive inflation. After inflation ends, the topological defects (e.g., domain walls)

produced by the decay of the tachyon field become the initial seeds for the formation of large-scale structure in the universe.

Baryon acoustic oscillations:

- *Stage 1 – Dark matter builds the skeleton first (end of inflation):* The high-dimensional tachyon field itself possesses tiny density inhomogeneities ($\delta\rho/\rho \sim 10^{-5}$). These small “balls” (dark matter candidates) do not interact electromagnetically with photons. This means that before the recombination epoch, when ordinary matter (baryons) is still “locked” by photon radiation pressure and cannot clump, the density perturbations of dark matter can already grow freely under gravity and collapse first, forming the “skeleton” (the cosmic web) of the large-scale structure of the universe – a network of filaments and gravitational potential wells.

- *Stage 2 – Ordinary matter fills the skeleton:* When the universe cools to recombination (about 380,000 years), photons decouple and ordinary matter (gas) becomes free. The gravitational potential wells (the skeleton) already built by dark matter begin to exert their influence. Ordinary matter (gas, dust) “falls” into these pre-existing potential wells under gravity. The ordinary matter further collapses and cools at the centres of the potential wells, forming galaxies and stars. Thus, the distribution of galaxies we see today is actually aligned along the dark-matter skeleton.

- *Late stage – Mutual locking (our universe holds onto the tachyons):* In the present universe (low redshift), structure has become highly nonlinear. **Reversal mechanism:** once ordinary matter has formed high-density galaxies and clusters (objects) in the dark-matter potential wells, these regions generate extremely strong local gravitational potentials. **Locking effect:** this deep gravitational potential well produced by the accumulation of ordinary matter further binds and gathers surrounding dark matter. This forms a positive gravitational feedback loop: dark matter pulls ordinary matter → ordinary matter accumulates and deepens the potential well → the well pulls in more dark matter.

Dark matter fluidity in the post-explosion era:

The projection of high-dimensional tachyons into three-dimensional space has dynamics determined by an effective potential:

$$V_{\text{eff}} \approx -\frac{GM}{r} + \frac{1}{2}m_{\text{eff}}v^2$$

If m_{eff} is large (the effective mass after compactification of the high-dimensional mass), it becomes “anchored” at the minimum of the gravitational potential (near the centres of galaxies). Therefore, in empty regions of our universe, dark matter flashes by at superluminal speeds; where the gravitational potential is strong, it is dragged and forms “shockwave” effects, leading to accumulation. This effective mass m_{eff} is not a free parameter. From the total-energy-zero condition and the stability radius $R_c = z_d/\alpha\ell_P$, the Schwarzschild relation $R_c = 2Gm_{\text{eff}}/c^2$ gives $m_{\text{eff}} = (z_d/2\alpha)M_P$. For $d=4,5,6$ and $\alpha \sim 1-10$, the mass lies in the narrow range **0.2–2.6 M_P** , i.e. $(0.4-5.7) \times 10^{-8}$ kg. This concrete prediction can be tested by direct detection and by the interferometric phase noise searches of Section 8. Crucially, this m_{eff} is not defined by integrating a density over open 3D space; instead, it characterizes a spatially segregated state—a localized patch with its own envelope, detached from the ambient 3D background. Consequently, the only external handle is its envelope scale R_c , linking

mass directly to geometry in a way that mirrors the Schwarzschild form but originates from the discrete structure of space itself.

Mysterious high-energy cosmic rays:

These are often referred to in cosmology as the “GeV/TeV gamma-ray excess”. It is commonly thought that dark matter annihilation is responsible. In regions of the universe with the strongest gravitational potentials (e.g., around supermassive black holes, centres of galaxies), the high-dimensional small “balls” (dark matter) are dragged by gravity and their density becomes extremely high. When two such balls come sufficiently close, according to quantum field theory, they can annihilate – i.e., the two particles collide and “disappear”, converting into standard-model particles (such as quarks, leptons) of our universe. These produced high-energy particles (e.g., quarks) are extremely unstable and quickly decay via strong interactions into a cascade of secondary particles, ultimately producing high-energy gamma rays. These signals have two characteristics: *Signal drowning*: the gamma rays produced by such annihilations are mixed with gamma rays from astrophysical processes such as black hole accretion disks and pulsars, making them very difficult to distinguish. *Extremely weak interaction*: the annihilation cross-section (probability of encounter) between high-dimensional balls is very small; even in the highest-density regions, annihilation events are rare.

7.5 Gravity and the Graviton

Gravity originates from the “mechanical-like” vibration of a closed space in response to its internal matter-energy. It is a necessity for the “spatial envelope” to maintain unitary evolution of its quantum state. The vibration consists of two parts: a radial vibration that cancels the energy of the internal matter, and an angular vibration that displays all the internal quantum information. The radial vibration of the gravitational surface is the necessary medium for achieving quantum entanglement. Gravity in the classical universe arises from the spontaneous curvature of otherwise flat space in order to achieve a low-energy state for a closed region, thereby maintaining the lowest-energy configuration of that region. The graviton is not a standard particle; the smallest manifestation of gravity is the spatial curvature energy caused by the projection of a high-dimensional tachyon into our universe. Its size is close to the Planck scale.

7.6 Quantum Entanglement

The essence of quantum entanglement is information exchange between spaces (angular information of the radial wavefunction). The mechanism for producing entanglement between low-energy, positive-energy matter in our universe is to allow information exchange of spatial phases between two particles in the extra dimensions. Due to the Pauli exclusion principle, two fermions are absolutely spatially separated without the participation of bosons; thus, they cannot directly become entangled. Bosons do not occupy space. Therefore, at the Planck scale, two bosons, or a fermion and a boson, can jointly “encode” an angular resonance in the extra dimensions through the angular vibrational information carried by the tachyon space.

7.7 The Illusion of Being at the Center of the Universe and Spatial Flatness

The illusion that we are at the centre of the universe arises because the true boundary of the universe lies far beyond the observable range. The “uniform” expansion of the universe stems from: (1) the geometric energy of flat space tends to zero, so tiny energy perturbations can stretch space; (2) high-dimensional tachyons continue to project into our universe under extreme conditions. The flatness of the observable universe: our universe (3D) is far larger than the observable universe; curvature could be observed at the decoherence wavefront (during inflation) only fleetingly. The cosmic microwave background radiation completely isolates our horizon from deeper cosmic structures.

7.8 Dark Energy

The geometric properties of space depend on its own shape. Flat space carries no geometric energy, while nearly flat space requires extremely little energy to create or annihilate. The observed stretching of space within the observable universe does not truly originate from the geometric energy of our three-dimensional cosmic space. The nature of dark energy depends on the **true** cosmic boundary far beyond the horizon.

7.9 Three-Dimensional Space in the Classical Universe

Why our universe appears three-dimensional can be understood from the relationship between geometric energy and dimension revealed by the radial spherical Bessel functions: the higher the dimension, the higher the geometric energy of space, making it easier to confine high-energy tachyons. In our lower-dimensional universe, particles adopt a “space-sharing” scheme that confines the binding energies of the strong and weak interactions to extremely small scales. As a three-dimensional sphere, the geometric energy of our universe is sufficient to independently handle long-range electromagnetic interactions, so there is no need to “split” space to maintain balance.

7.10 Particle Identity (Why Particles Are Identical)

From the diverse, high-dimensional background (tachyon sea) to identical particles in our low-dimensional projection (our universe): in the high-dimensional network of the tachyon sea, there is no unified concept of “space”. The vibrational modes, imaginary masses, and phases of each “tachyon” (node) can be independent and different. Between them there are only quantum correlations, no spatial distances. Therefore, in their native state, they are not identical; they are chaotic “quantum information”. When these high-dimensional nodes are “woven” into our 3+1 dimensional spacetime (spatial envelopes) via a topological phase transition, a fundamental change occurs: **Space sharing**: this “shared space” possesses Lorentz symmetry (rotational and translational invariance). **Symmetry enforcement**: to satisfy the symmetry of this macroscopic space, particles must “disguise” themselves as linear excitations of this space. The energies of particles are classified and interact according to symmetry, and eventually they become identical within this “shared space”. The origin of the Pauli exclusion principle: space is a finite resource. In the low-dimensional projection, position eigenstates form a shared basis. If two fermions attempt to occupy exactly the same spatial quantum state, this would correspond to a superposition error in the low-dimensional projection (high-dimensional information cannot be uniquely mapped), and the system forcibly excludes such ambiguity, manifesting as a repulsive force.

7.11 Intrinsic Properties of Particles and the Cosmological Constant

The intrinsic symmetries of particles in the classical universe (such as charge, colour, flavour, etc.) originate, in essence, from the collapse of rotational symmetries in higher dimensions (or the “tachyon-space-envelope duality”) into the shared three-dimensional space. Because these symmetries can no longer “resonate” with the external space, they become internalised as intrinsic degrees of freedom (Kaluza-Klein theory, T-duality). The $U(1)$ symmetry inside a tachyon inverts and “re-materialises” as the continuous electromagnetic interaction in the external space of the classical universe, and this force cancels with the gravity of the shared space. The high-dimensional rotational origins of the discrete symmetries of the Standard Model, $SU(3) \times SU(2)$, are $SO(6) \times SO(4)$, suggesting that the optimal dimensions for tachyons are likely 4 and 6. The spinor covering group of $SO(4)$, $Spin(4)$, is isomorphic to $SU(2) \times SU(2)$. Under $SO(4) \rightarrow SU(2)$: the rotational symmetry of a 4-dimensional hypersphere, in its spinor representation, splits into two $SU(2)$ factors, one of which corresponds to the weak force. Similarly, $SO(6) \rightarrow SU(3)$: the rotational symmetry of a 6-dimensional hypersphere, in its spinor representation, splits into $SU(4)$, and its projection into three-dimensional space yields $SU(3)$.

Core hypothesis: gauge interactions originate entirely from the geometric structure of the high-dimensional tachyon space. Specifically:

- The weak force ($SU(2)$ symmetry) originates from the “flipped projection” onto three-dimensional space of angular vibrational modes on the surface of a 4-dimensional sphere (S^3).
- The strong force ($SU(3)$ symmetry) originates from the “flipped projection” onto three-dimensional space of angular vibrational modes on the surface of a 6-dimensional sphere (S^5).

The strength (coupling constant) of each force is determined by the critical radius of its corresponding high-dimensional sphere. The critical radius is the characteristic scale of stable vibrational modes on the high-dimensional sphere, given by the first zero of the zero-angular-momentum radial Bessel function for that dimension.

Obtaining the critical radii (in natural Planck units):

- Critical radius for the 4-dimensional tachyon sphere (corresponding to the weak force): $R_w = 3.831706$
- Critical radius for the 6-dimensional tachyon sphere (corresponding to the strong force): $R_s = 5.135622$

Establishing a scaling relation between force strength and radius: In Kaluza-Klein type theories, the gauge coupling constant g is typically proportional to some power of the compactification scale. Based on dimensional analysis and geometric intuition (e.g., force may be related to curvature or energy density, and energy density $\rho \sim 1/R^2$, but the coupling constant might be proportional to the square of the characteristic scale), we make the simplest assumption: **the gauge coupling constant g is proportional to the square of the corresponding critical radius**, i.e., $g \propto R^2$. The physical picture behind this assumption is: a larger critical radius corresponds to higher energy of the vibrational mode, and in the low-energy effective theory, the interaction appears stronger.

Calculating the ratio: Under the above assumption, the ratio of the strong coupling constant to the weak coupling constant is

$$\frac{g_s}{g_w} = \left(\frac{R_s}{R_w}\right)^2 = \left(\frac{5.135622}{3.831706}\right)^2 \approx 1.7965$$

Comparison with experimental values: At low energy scales (e.g., the QCD scale and the electroweak scale), the experimentally measured coupling constants are approximately $g_s \approx 1.22$ and $g_w \approx 0.65$, giving a ratio

$$\frac{g_s}{g_w} \approx \frac{1.2}{0.65} \approx 1.8462$$

The relative error between the theoretical prediction and the experimental value is about 2.7%. Possible sources of error include projection effects from the higher dimensions and curvature effects.

7.12 Realisation of the Holographic Principle:

In the tachyon sea model, the holographic principle is manifested as a deep duality between matter and geometry: for a stably existing tachyon, the internal perspective is a high-energy, imaginary-mass string vibration (language of particle physics / string theory), while the external perspective is the violent oscillation of the high-curvature, negative-energy geometric envelope that contains it (language of loop quantum gravity). The two are inseparable aspects of the same physical reality – the vibrational information is fully encoded in the angular momentum modes of the higher-dimensional brane. The high-dimensional tachyon sea is naturally in a state of global entanglement; with its high dimensionality and numerous connections, the concepts of “distance” and “separation” are almost absent, requiring no additional bridges. When our own universe emerges as a low-dimensional projection / slice of the tachyon sea, the rich high-dimensional entanglement is compressed into locality and causality. Consequently, from the low-dimensional perspective, the intrinsic entanglement connections must be described geometrically via wormholes (ER bridges) – this is precisely how ER = EPR is realised in the present model: the ER bridge is the projected geometric interpretation of the inherent quantum entanglement in the high-dimensional tachyon sea, as seen from the low-dimensional universe.

7.13 The Destiny of the Universe – An Eternal Cyclic Symphony

The mechanisms by which matter in our universe returns to the tachyon sea are twofold: **Density compression** – e.g., via black holes. **Density dilution** – when a region (such as a cosmic void) expands to the point that its horizon size becomes smaller than the void, the matter density becomes too low to sustain the geometry of that region, causing it to lose causal contact with the outside. The matter is then forced back into an imaginary-mass tachyon state. Cosmic voids and the cosmic web together compose the grand evolutionary narrative of large-scale structure. As the horizon gradually shrinks within a void, the particles inside eventually lose causal connection with the surrounding space. When the matter density can no longer sustain the geometric energy of that region, space undergoes a flip and compression, driving all matter into an extremely high-density state, ultimately achieving the reversal from real to imaginary mass and returning to the tachyon sea.

Meanwhile, regions of the cosmic web continue to contract, causing matter density to increase; once a critical threshold is exceeded, black holes form. Black holes continuously devour surrounding matter, likewise sending it back to the tachyon sea. After our universe fades away, the next universe is born through a decoherence process within the tachyon sea, beginning a new round of evolution, dissolution, and return – an eternal cyclic symphony.

8. Verification Experiment Proposal: Detecting Non-Gaussian Interference Signatures of High-Dimensional Phase Projections

8.1 Core Principle

In the tachyon-sea model, high-dimensional tachyons stabilized on four-dimensional and six-dimensional spheres project their angular information into our three-dimensional space without transferring energy. This projection imprints a characteristic non-Gaussian phase modulation onto ordinary matter, such as atoms or light. The modulation carries the topological signature of the original higher-dimensional sphere, and its angular distribution on the sky can only be decomposed into the eigenmodes of the spherical harmonics of the four-sphere and the six-sphere. Ultra-deep underground laboratories provide an environment of exceptionally low seismic, thermal and gravitational-gradient noise, making them the ideal locations for long-baseline, long-integration phase measurements.

8.2 Optimal Experimental Platform: Underground Long-Baseline Atom Interferometers

Conventional dark matter detectors such as LZ, XENONnT, PandaX and DAMA/LIBRA rely on energy deposition, including nuclear or electron recoil, scintillation, or heat. These detectors are completely blind to zero-energy, pure-phase signals. In contrast, atom interferometers directly measure the phase of a matter wave as it propagates through a long vacuum tube. The atoms, typically rubidium-87 or potassium-40, are split, reflected and recombined by laser pulses, and any tiny phase shift accumulated along the path is read out via the final atomic population. Several underground facilities of this type are either in operation, under construction, or in advanced planning. MAGIS-100 at Fermilab in the United States is a 100-meter baseline atom interferometer serving as a prototype for gravitational wave and ultralight dark matter detection. MIGA at the LSBB underground laboratory in Rustrel, France is a large-scale hybrid atom-laser antenna embedded in a low-noise environment, using a 200-meter optical cavity to simultaneously interrogate multiple atomic ensembles. ZAIGA is a proposed underground atom interferometer gravitation antenna near Wuhan, China, with a baseline of several hundred meters, located on average 200 meters below the surface of Zhaoshan mountain. The low-noise underground setting suppresses environmental disturbances such as seismic, thermal and gravity gradient noise, and allows for extremely long integration times, from months to years. This is essential for extracting a weak, persistent non-Gaussian phase signal.

8.3 Observables and Data Analysis

For each pointing direction of the interferometer baseline relative to the celestial sphere, the apparatus records a time series of the differential phase. After subtracting all known contributions

from seismic noise, laser frequency noise, gravity gradient noise, thermal noise and others, the residual phase contains the candidate signal together with remaining noise. Because ordinary noise sources are Gaussian, their higher-order cumulants vanish. The predicted tachyon projection, in contrast, is non-Gaussian and locked to the sky direction. Therefore the analysis proceeds in two steps. First, one computes the bispectrum, which is a third-order quantity, and the trispectrum, a fourth-order quantity, of the residual phase. Any persistent non-zero higher-order cumulant that varies with pointing direction is a candidate signal. Second, one decomposes the spatial map of the residual phase variance or bispectrum amplitude into spherical harmonic components. Ordinary three-dimensional spherical harmonics are used as a baseline, but the hypothesis is that the signal is better described by a finite set of projected eigenmodes from the four-sphere and the six-sphere. The predicted dominant projected multipoles are the dipole and quadrupole from the four-sphere, and the quadrupole, octupole and hexadecapole from the six-sphere. If the tachyon-sea model is correct, the coefficients associated with the higher-dimensional eigenmodes are significantly non-zero, while the ordinary spherical harmonic coefficients are consistent with zero after appropriate cleaning.

8.4 Immediate Low-Cost Pilot: Reanalysis of Existing Data

The most cost-effective first step is to reanalyse already collected or near-term data. MAGIS-100, even in its commissioning phase, continuously records atom interferometer phase time series with known pointing information. These data can be searched for the predicted non-Gaussian, sky-correlated residuals using the higher-order cumulant method described above. For laser interferometers such as LIGO, Virgo and KAGRA, the publicly available auxiliary channels, including DARM, MICH and alignment sensor phase time series, also contain long, continuous phase records with known detector orientation. Although these are optical rather than atomic interferometers, they can be analysed for the same type of non-Gaussian phase correlations. A positive detection in existing data would be a groundbreaking validation of the tachyon-sea model. A null result, once sufficient sensitivity is reached, would place meaningful upper bounds on the coupling between higher-dimensional tachyons and our three-dimensional space.

8.5 Distinguishing from Known Physics

Seismic and thermal noise are Gaussian and have no sky-locked angular structure, and their higher-order cumulants vanish. Gravitational waves produce an amplitude signal, not a pure phase signal, and are primarily quadrupolar with no bispectrum or trispectrum in vacuum. Laser frequency noise is common-mode in the two arms and can be subtracted. Conventional dark matter candidates such as WIMPs or axions produce either energy deposition or a narrow-band spectral line, not a zero-energy, non-Gaussian, broadband phase correlation. The cosmic microwave background has no mechanism to imprint a phase in a laboratory interferometer. Therefore the predicted signature is unique to the tachyon projection mechanism.

8.6 Interpretation of a Positive Result

A detected signal consistent with the projected four-sphere and six-sphere harmonic modes would provide the first experimental evidence for the existence of a tachyon sea with imaginary mass and higher-dimensional stability. It would validate the duality between imaginary-mass

string-theory excitations and the oscillating geometric envelope of loop quantum gravity. It would reveal a new, zero-energy, topology-locked quantum correlation, offering a direct observational window into quantum gravity without requiring Planck-scale colliders. Finally, it would provide a concrete laboratory test for the geometric origin of gauge symmetries, with the weak force coming from the four-sphere and the strong force from the six-sphere. A null result after achieving the target sensitivity would place stringent constraints on the tachyon-matter coupling and rule out the simplest version of the model.

Solving the tachyon equations exactly and applying T-duality leads to a derivation of the cosmological constant – this can also be regarded as a key success of the theory.

9. Conclusion

In summary, the tachyon-sea decoherence cosmological model proposed in this paper successfully eliminates the Big Bang singularity, unifies relativity and quantum mechanics, and bridges the dual descriptions of string theory and loop quantum gravity within a single physical picture. The model not only fully reproduces the standard evolutionary course after cosmic inflation, but also provides an internally consistent causal mechanism for a long list of long-standing open problems: the origin of spacetime, the black hole information paradox, the nature of gravity, dark matter, vacuum energy, the smallness of the cosmological constant, the origin of quantum entanglement, the emergence of three-dimensional space, the identity of particles, and more. The entire universe exhibits a cyclic, renormalising dynamic process, with total energy zero, and gravity and quantum field theory naturally coupled in the dual framework of tachyons and geometric envelopes. The work presented here suggests that the “fully entangled tachyon sea” can serve as a concrete, testable starting point toward a unified theory of quantum information and geometry.

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The conceptual model detailed here is the foundational work of the authors. Any research that produces a mathematical formalization or discovers new physical implications based on this framework must provide proper citation to this original work. The authors express an interest in collaborative development and request that any derivative work, at a minimum, includes an appropriate citation or acknowledgement of this article's authors

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